

BITUNG, Indonesia

WMO: 970160

Lat: 1.4431N

Lon: 125.1800E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	23.8	19.5	14.3	27.5	20.3	15.0	27.7	7.8	28.3	6.4	28.3	0.2	160	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.0	33.4	25.7	33.0	25.7	32.6	25.7	27.1	31.4	26.7	31.1	26.6	31.0	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	29.7	25.6	20.8	29.5	25.2	20.4	29.3	85.3	31.6	84.2	31.5	83.2	31.3	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.1	4.3	21.4	35.4	1.5	1.6	20.3	36.5	19.4	37.5	18.6	38.4	17.5	39.6		
(5)			19.6	28.2	1.2	0.7	18.8	28.8	18.0	29.2	17.4	29.6	16.5	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	28.5	28.2	28.3	28.1	28.2	27.9	27.7	27.8	28.0	28.2	28.3	28.6		
	DBStd	0.90	0.98	0.98	0.89	0.90	0.88	0.99	0.80	0.79	0.77	0.79	0.71	0.85		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6624	574	511	569	543	563	537	548	551	539	563	549	577		
	CDD18.3	3582	316	278	311	293	305	287	290	292	289	305	299	318		
	CDH23.3	40079	3618	3108	3508	3281	3481	3152	3098	3118	3197	3449	3364	3703		
	CDH26.7	14226	1389	1171	1355	1218	1254	1008	870	866	1031	1270	1313	1481		
(14) Wind	WSAvg	1.4	1.6	1.5	1.3	1.1	1.1	1.3	1.6	2.1	1.7	1.1	0.9	1.3		
(15) Precipitation	PrecAvg	2518	323	263	244	213	230	207	133	94	95	171	265	270		
	PrecMax	3591	611	549	443	359	342	362	333	294	237	338	458	396		
	PrecMin	1553	110	55	33	49	93	0	0	0	2	19	158	157		
	PrecStd	472	116	114	116	79	63	81	90	82	66	84	74	71		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.6	33.8	33.5	32.7	32.4	31.9	31.9	32.3	32.9	33.5	33.7		
		MCWB	25.6	24.9	25.2	25.7	26.5	26.4	26.1	25.9	26.0	26.1	26.2	26.0		
	2%	DB	33.0	32.9	33.1	32.7	32.2	31.7	31.1	31.1	31.6	32.2	32.8	33.2		
		MCWB	25.6	25.2	25.2	25.9	26.4	26.1	25.8	25.7	25.8	26.0	26.1	26.0		
	5%	DB	32.3	32.2	32.5	32.0	31.6	31.1	30.6	30.5	31.1	31.8	32.2	32.6		
		MCWB	25.5	25.2	25.2	26.0	26.2	25.9	25.6	25.4	25.6	25.9	26.1	25.9		
	10%	DB	31.7	31.6	31.8	31.3	31.0	30.4	30.0	29.9	30.5	31.2	31.7	31.9		
		MCWB	25.4	25.2	25.3	26.0	26.1	25.7	25.4	25.2	25.4	25.8	26.0	25.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.0	26.7	27.7	27.2	26.9	26.7	26.6	26.7	27.1	27.2	27.2		
		MCDB	31.8	31.4	31.4	31.2	31.4	31.4	30.7	30.6	31.1	31.5	31.8	32.0		
	2%	WB	26.5	26.3	26.3	27.0	26.8	26.6	26.2	26.1	26.2	26.6	26.7	26.7		
		MCDB	31.4	30.9	31.0	30.9	30.9	30.7	30.2	30.1	30.5	31.0	31.3	31.6		
	5%	WB	26.1	26.0	26.0	26.6	26.6	26.2	25.9	25.7	26.0	26.3	26.4	26.4		
		MCDB	30.9	30.6	30.7	30.5	30.6	30.1	29.8	29.6	30.1	30.5	31.1	31.2		
	10%	WB	25.8	25.6	25.7	26.2	26.3	26.0	25.6	25.5	25.7	26.0	26.2	26.1		
		MCDB	30.3	30.1	30.2	30.2	30.2	29.7	29.3	29.2	29.6	30.3	30.8	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.8	7.1	6.8	6.1	5.3	4.9	4.7	5.6	6.7	7.5	7.0		
		MCDBR	7.5	7.6	8.1	7.7	6.8	6.1	5.6	5.4	6.1	7.2	8.1	7.8		
	5% WB	MCWBR	2.7	2.7	2.7	3.0	2.8	2.6	2.6	2.5	2.8	3.1	3.1	2.8		
		MCWBR	7.1	7.2	7.3	7.0	6.4	5.7	5.4	5.1	5.8	6.9	7.7	7.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.419	0.424	0.428	0.428	0.423	0.420	0.418	0.427	0.438	0.479	0.442	0.426		
		taud	2.489	2.464	2.452	2.443	2.465	2.485	2.498	2.456	2.410	2.282	2.395	2.459		
	Ebn at Noon	Ebn at Noon	909	908	898	879	861	851	859	868	876	848	880	897		
		Edh at Noon	113	117	119	117	110	106	106	114	122	139	123	115		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.00	5.41	5.73	5.66	5.45	5.03	5.33	5.76	6.10	5.91	5.48	5.15		
	RadStd	RadStd	0.29	0.45	0.46	0.41	0.34	0.33	0.47	0.35	0.40	0.35	0.32	0.36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.30	N/A	N/A	+0.94	+0.63	+0.45	N/A	N/A	N/A	N/A

Nomenclature: See separate page